

Short CV

José Ferreira

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EDUCATION

- PhD in Physics** 08/2023 - PRESENT
University of Aveiro
Supervisors: Miguel Zilhão, Carlos Herdeiro and Raimon Luna
- MSc in Physics** 09/2020 - 09/2022
Faculty of Sciences of the University of Lisbon
Thesis: Constraining $f(Q)$ Cosmology with Standard Sirens
Supervisors: Nelson Nunes and Tiago Barreiro
- BSc in Physics** 09/2017 - 07/2020
Faculty of Sciences of the University of Lisbon

RESEARCH INSTITUTES

- Gr@v** 08/2023 - PRESENT
Member of the Gravitation group integrated in the Center for Research & Development in Mathematics and Applications (CIDMA) of the University of Aveiro, Portugal.
- Institute of Astrophysics and Space Sciences** 09/2021 - 07/2023
Member of the cosmology group at the Institute of Astrophysics and Space Sciences, Lisbon node, Portugal.

TEACHING

- Assistant** 09/2022 - PRESENT
University of Aveiro
Courses: Numerical and Statistical Methods
- Assistant** 02/2022 - 07/2023
Faculty of Sciences of the University of Lisbon
Courses: Experimental Physics I, Elements of Physics, Mathematical Methods for Physicists, General Physics

FELLOWSHIPS

- PhD scholarship** 01/2024 - PRESENT
Portuguese National Foundation for Science and Technology
A PhD scholarship to perform numerical evolution of hairy black holes in General Relativity.
- Research Fellowship** 08/2023 - 01/2024
Gr@v
A fellowship to study hairy black holes, properties and stability, in General Relativity.
- Research Fellowship** 02/2023 - 07/2023
Institute of Astrophysics and Space Sciences
A fellowship to study cosmological models based on non-metric gravity using current and forecast gravitational wave events.

PUBLICATIONS

- José Ferreira, Tiago Barreiro, José Mimoso, and Nelson J. Nunes. “Testing Λ -free $f(Q)$ cosmology”. *Phys. Rev. D* 108 (6 Sept. 2023), p. 063521. doi: [10.1103/PhysRevD.108.063521](https://doi.org/10.1103/PhysRevD.108.063521).
- José Ferreira, Tiago Barreiro, José Mimoso, and Nelson J. Nunes. “Forecasting $F(Q)$ cosmology with Λ CDM background using standard sirens”. *Phys. Rev. D* 105 (12 June 2022), p. 123531. doi: [10.1103/PhysRevD.105.123531](https://doi.org/10.1103/PhysRevD.105.123531).



COMMUNICATIONS

CONTRIBUTED TALKS

“Splitting the (Gravitational) Atom”. 21th-25th of July, 2025. New Frontiers in NR 2025, Mallorca, Spain. (Links: [pptx](#), [pdf](#))

“(In)stability of Q-Hairy Black Holes”. 20th-22th of May, 2025. NewFunFICO Meeting, Valencia, Spain. (Links: [pptx](#))

“Non-linear Superradiance of Charged Black Holes: Beyond Spherical Symmetry”. 18th-22th of November, 2024. VI Amazonian Symposium on Physics, Belém, Brazil. (Links: [pptx](#), [pdf](#))

“Using Standard Sirens to test non-metric theories of gravity”. 19th-20th of December, 2023. XVI Black Holes Workshop, Porto, Portugal. (Links: [pptx](#), [pdf](#))

SEMINARS

“Cosmological Tests on Non-Metricity Based Theories of Gravity”. 20th of September, 2023. Gr@v, Aveiro, Portugal. (Links: [recording](#), [pptx](#), [pdf](#))

POSTERS

“Splitting the (Gravitational) Atom”. 21th-25th of July, 2025. New Frontiers in NR 2025, Mallorca, Spain. (Links: [pdf](#))

“Testing Λ -Free $f(Q)$ Cosmology”. 3th-5th of May, 2023. Dark Matter and Stars, Lisbon, Portugal. (Links: [pdf](#))



DIGITAL SKILLS

High Level Languages Julia, Python, Mathematica

Multimedia Editing GIMP, Photoshop, Inkscape, Kdenlive, Audacity

Low Level Languages C, Fortran, CUDA

Web Development Quarto, Hugo

I consider myself an advanced Linux user, with experience using it on personal machines, servers, clusters and microcontrollers. I work mostly on a terminal based ecosystem, creating and making use of terminal utilities to develop my projects.



ACADEMIC ACTIVITIES

Supervisor at the IAstro Summer Internships 2023

07/2023

Institute of Astrophysics and Space Sciences

Supervisor of two undergraduate projects that consisted on constraining modified gravity cosmological models with current and forecast datasets coming from different sources.

Podcast “Physically Speaking”

12/2020 - 12/2022

[Fisicamente Falando](#) [Fisicamente Falando](#) [fisicamentefalando.nfef](#) [fisicamente_falando_](#)

A student driven podcast created during the pandemic where we interviewed faculty members to speak about their lives, both inside and outside of academia. I am in charge of the audio and video editing and also take part in some of the interviews.

Member of the Student Body for Physicists and Physical Engineers

10/2020 - 10/2021

[nfef-fcul.com](#) [NFEFFCUL](#) [nfeffcul](#)

I was in charge for the development of the digital tools of the student body of physicists and physical engineers. This included the website, the online repository (which is made up of student provided content) and the Discord channel.



OUTREACH

“Echoes and Shadows: How to Detect Black Holes”

03/05/2024

[Presentation \(.pptx\)](#)

A Presentation, for high school students, on what are black holes and how to find them.

European Researchers’ Night 2024

27/09/2024

Coordinator of the participation of Gr@v in the European Researcher’s Night in Aveiro, Portugal. Using digital content, we explained the physics behind Gravitational Waves and lensing around compact objects.

“When a Black Hole Falls Inside... Another Black Hole”

15/04/2023

[Presentation \(.pptx\)](#)

Presentation of an Ignite style talk in an event organized by the Institute of Astrophysics and Space Sciences that brings space sciences to cities far away from the main urban areas.